Do Now · retrieval

Lesson 135 · Entity Relationships: One-to-Many and Many-to-Many

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the role of a foreign key?

Q2

In Big O notation, what is binary search worst-case complexity?

Q3

What does ROM stand for?

Q4

Which is removed first from a FIFO queue?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify entities — Find objects with independent identity, not every noun.

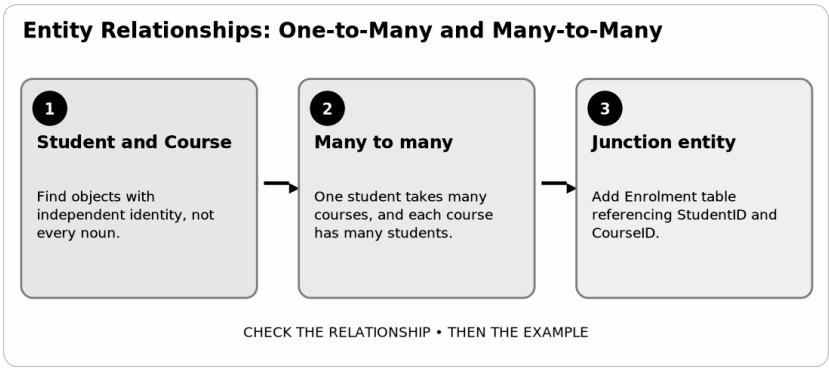
Core idea 2

Set cardinality — One student takes many courses, and each course has many students.

Core idea 3

Resolve M:N — Add Enrolment table referencing StudentID and CourseID.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

entity — An identifiable class of thing stored in database.

relationship — Association between entities.

cardinality — How many related instances are possible.

junction table — Table breaking a many-to-many relation into two one-to-many relations.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Represent students enrolling on several courses, with many students per course.

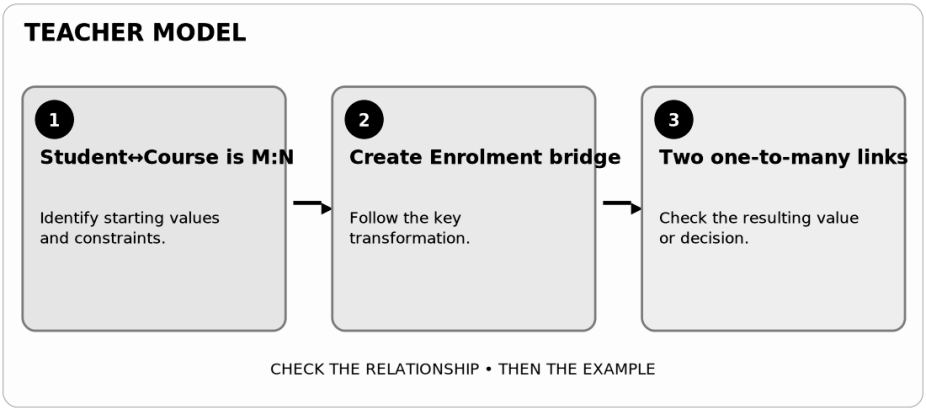
Follow the worked method

1. Student↔Course is M:N — Find the given data and name the required result.

2. Create Enrolment bridge — Explain why this intermediate stage is necessary.

3. Two one-to-many links — Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

One Customer
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the difference between one-to-many and many-to-many relations with examples. [6]

Question 2

Design tables for Author and Book when each book has multiple authors. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate why a field containing comma-separated ClubIDs is poorer than a Membership junction table. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A many-to-many relationship can always be represented correctly with just a single FK on one of the two entity tables.

Use key vocabulary from memory

1. Explain entity in your own words.

2. Explain relationship in your own words.

3. Explain cardinality in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.